

Student Tim Greß (12412672)
Supervision Main: Sebastian Hattinger (WU Wien); Univ.-Prof. Dr. Emanuel Sallinger (TU Wien)
Lecture Foundations of International Business (WU) – parallel to project

Global Player Mobility in Football: A League-to-League Network Analysis

Project Proposal

Abstract

This project analyzes global player mobility in professional football using transfer data from the Transfermarkt website. Player moves are represented as a directed, weighted network on league level, where edges capture transfer flows between source and destination competitions across seasons. Using standard network analysis methods (centrality measures, community detection, and tier-flow indicators), the project aims to describe league roles such as exporters, importers, and bridging markets, and to identify mobility regions and lower- to top-division pathways. The analysis follows a CRISP-DM workflow and evaluates the results through hypothesis testing, robustness checks, and comparison to a simple randomized baseline network.

Motivation and Problem Statement

Professional football can be seen as a global labor market, where players move between clubs and leagues across countries. Many commonly used terms such as “selling leagues”, “import leagues”, or “stepping-stone pathways” are mostly qualitative and not easy to reproduce. A league-to-league network gives a clear and data-driven view of mobility pipelines and interconnections across seasons. The dataset from Transfermarkt contains structured information on players, clubs, leagues and transfers (including transfer type and fees), so it should be possible to analyze movements between countries and also between lower and top divisions.

Research Questions

- **RQ1 – Mobility structure / concentration:** How is player mobility between leagues structured as a directed and weighted network, and how concentrated are incoming transfer flows across leagues?
- **RQ2 – Centrality and roles:** Which leagues primarily act as senders or receivers of players, and which leagues function as bridges between otherwise weakly connected parts of the network?
- **RQ3 – Communities / mobility regions:** Are there clusters of leagues that form strongly connected mobility ecosystems and how stable are these clusters across seasons?
- **RQ4 – Lower-league pathways:** How important are transfers between lower tiers and top tiers, and how do these pathways differ from top-tier mobility?

Hypotheses

- **H1 – Concentration:** Incoming transfer flows are highly concentrated, with a small set of leagues accounting for a large share of all incoming transfers.
- **H2 – Bridges:** Some mid-level leagues exhibit disproportionately high betweenness centrality and act as bridges between regions and/or tier systems.
- **H3 – Communities:** The network exhibits non-random community structure, with detected communities showing higher modularity and/or greater stability than in a randomized baseline network.
- **H4 – Lower-league pathways:** Transfers originating from lower tiers are more likely to be domestic and upward (from lower to higher tier) within the same country, while cross-border moves are relatively less common compared to top-tier mobility.

Methodology

The project follows CRISP-DM and standard data science practice.

- **Business understanding:** Define scope (years, competitions/leagues, and which transfer types to include). This is mainly determined by the available data.
- **Data understanding:** Inspect coverage, missing values, and consistency of league/club identifiers and seasons. This was already done on a small subset to write this proposal.
- **Data preparation:** Map each transfer to *source league* → *destination league* for a given season. Create edge lists per season and an aggregated graph. Include loans as an attribute. Store fee information as an attribute.
- **Modeling / analysis:** Build directed weighted graphs per season and overall. Compute (i) import/export rankings (weighted in/out degree), (ii) betweenness for bridge leagues, and (iii) community detection (Louvain/Leiden). Report domestic vs. international shares and tier-flow indicators (lower→top division).
- **Evaluation:** (i) Test the hypotheses using the computed metrics and clear summaries, (ii) robustness checks (with vs. without loans; season-wise vs. aggregated), and (iii) baseline comparison using a simple null model (randomized graph with preserved degrees) to show whether the observed structure is stronger than random.

Expected Results and KPIs / Success Criteria

- **Reproducible pipeline:** End-to-end notebook that builds the league mobility graphs per season from raw data.
- **Core outputs per season:** (a) summary table with top importers/exporters and bridge leagues, (b) community assignment, (c) domestic vs. international share.
- **Visual outputs:** Clear figures: league-to-league flow heatmap and community/network visualizations.
- **Null model result:** Report at a quantitative comparison vs. the randomized baseline (e.g. higher modularity or stronger flow concentration than the null).